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Jensen's Inequality for Operators and Löwner's Theorem

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1. Introduction

In 1905 Jensen showed [8, 9] that a (mid-point) convex, continuous function f on an interval I satisfies

$$\text{(J)} \qquad f(\Sigma\lambda_i t_i) \leqq \Sigma\lambda_i f(t_i)$$

for any convex combination $\{\lambda_i\}$ of points $\{t_i\}$ in I. Taking

$$a = \begin{pmatrix} \lambda_1^{1/2} & 0\ldots 0 \\ \vdots & \vdots \quad \vdots \\ \lambda_n^{1/2} & 0\ldots 0 \end{pmatrix}, \qquad x = \begin{pmatrix} t_1 & & 0 \\ & \ddots & \\ 0 & & t_n \end{pmatrix},$$

and considering f as a function on self-adjoint matrices [with $f(0)=0$], the inequality (J) reads

$$\text{(JO)} \qquad f(a^*xa) \leqq a^* f(x) a.$$

We shall say that a continuous, real function f on I satisfies *Jensen's Operator Inequality* if (JO) holds for any self-adjoint matrix x with spectrum in I and every matrix a with $\|a\| \leqq 1$. Since we put no bound on the order of x and a, and the finite rank operators are dense in $B(H)$, H a Hilbert space, this is equivalent to demanding (JO) for any self-adjoint x in $B(H)$ with spectrum in I and every a in $B(H)$ with $\|a\| \leqq 1$.

A technicality comes in here. Since zero may be an eigenvalue of a (indeed, $a=0$ is not excluded), zero may belong to the spectrum of a^*xa. Thus the interval I must contain zero in order for (JO) to be meaningful. Moreover, the element a^*xa really is a non-commutative generalization of a *degenerate* convex combination of a and 0 (i.e. $\Sigma\lambda_i \leqq 1$). The "correct" generalization of a convex combination of x and y would seem to be an operator $a^*xa + b^*yb$, where $a^*a + b^*b = 1$ [cf. 2.1(iii)]. But correctness must bow to applicability, so we shall stick to (JO). Anyway it is clear that a convex function f will satisfy Jensen's inequality (J) for all degenerate convex combinations, precisely when $0 \in I$ and $f(0) \leqq 0$.

As usual we shall say that a real function f on an interval I (of any type) is *operator monotone* if for each n and every pair x, y of self-adjoint elements in the

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algebra $\mathbb{M}_n$ of $n\times n$-matrices with spectra in I, the condition $x\leqq y$ implies $f(x)\leqq f(y)$. Likewise we say that f is *operator convex* on I if for each n in $\mathbb{N}$

$$f(\lambda x+(1-\lambda)y)\leqq\lambda f(x)+(1-\lambda)f(y)$$

for all self-adjoint x,y in $\mathbb{M}_n$ with spectra in I and every λ in $[0,1]$. From this definition it is clear that a pointwise limit of operator monotone (respectively operator convex) functions is again operator monotone (respectively operator convex). A full account of the theory of these functions is told in [6]. Short but concise surveys are found in [5] and [1]. A short proof of Löwner's theorem appeared in [14] after this work was completed. There is no overlap with our proof, which is oriented towards operator algebra theory, where operator monotone and operator convex functions have found very useful applications, see e.g. [13, 1.3].

2. Jensen's Operator Inequality

2.1. Theorem. *If f is a continuous, real function on the half-open interval $[0,\alpha[$ (with $\alpha\leqq\infty$), the following conditions are equivalent:*

(i) *f is operator convex and $f(0)\leqq 0$.*

(ii) *$f(a^*xa)\leqq a^*f(x)a$ for all a with $\|a\|\leqq 1$ and every self-adjoint x with spectrum in $[0,\alpha[$.*

(iii) *$f(a^*xa+b^*yb)\leqq a^*f(x)a+b^*f(y)b$ for all a,b with $a^*a+b^*b\leqq 1$ and all x,y with spectra in $[0,\alpha[$.*

(iv) *$f(pxp)\leqq pf(x)p$ for every projection p and every self-adjoint x with spectrum in $[0,\alpha[$.*

Proof. (1)⇒(ii): Consider the operators on $H\oplus H$ given by

$$X=\begin{pmatrix}x&0\\0&0\end{pmatrix},\qquad U=\begin{pmatrix}a&b\\c&-a^*\end{pmatrix},\qquad V=\begin{pmatrix}a&-b\\c&a^*\end{pmatrix},$$

where $c=(1-a^*a)^{1/2}$ and $b=(1-aa^*)^{1/2}$. Calculation shows that U and V are unitaries and that

$$U^*XU=\begin{pmatrix}a^*xa&a^*xb\\bxa&bxb\end{pmatrix},\qquad V^*XV=\begin{pmatrix}a^*xa&-a^*xb\\-bxa&bxb\end{pmatrix}.$$

Since f is operator convex we obtain

$$\begin{aligned}\begin{pmatrix}f(a^*xa)&0\\0&f(bxb)\end{pmatrix}&=f\begin{pmatrix}a^*xa&0\\0&bxb\end{pmatrix}\\&=f(\tfrac12U^*XU+\tfrac12V^*XV)\leqq\tfrac12f(U^*XU)+\tfrac12f(V^*XV)\\&=\tfrac12U^*f(X)U+\tfrac12V^*f(X)V\\&=\tfrac12U^*\begin{pmatrix}f(x)&0\\0&f(0)\end{pmatrix}U+\tfrac12V^*\begin{pmatrix}f(x)&0\\0&f(0)\end{pmatrix}V\\&\leqq\tfrac12U^*\begin{pmatrix}f(x)&0\\0&0\end{pmatrix}U+\tfrac12V^*\begin{pmatrix}f(x)&0\\0&0\end{pmatrix}V\\&=\begin{pmatrix}a^*f(x)a&0\\0&bf(x)b\end{pmatrix}.\end{aligned}$$

In particular, $f(a^*xa)\leqq a^*f(x)a$.

(ii)⇒(iii): Given a, b, x, and y as specified, put

$$A=\begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix}, \quad X=\begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix}.$$

Then $A^*A \leqq 1$ and $\mathrm{Sp}(X) \subset [0, \alpha[$ so that

$$\begin{aligned} \begin{pmatrix} f(a^*xa+b^*yb) & 0 \\ 0 & f(0) \end{pmatrix} &= f(A^*XA) \\ &\leqq A^*f(X)A = \begin{pmatrix} a^*f(x)a+b^*f(y)b & 0 \\ 0 & 0 \end{pmatrix}, \end{aligned}$$

as desired.

(iii)⇒(iv) is obvious.

(iv)⇒(i): Given operators x, y with spectra in $[0, \alpha[$ and λ in $[0, 1]$, define

$$X=\begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix}, \quad U=\begin{pmatrix} \lambda^{1/2} & -(1-\lambda)^{1/2} \\ (1-\lambda)^{1/2} & \lambda^{1/2} \end{pmatrix}, \quad P=\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Then $\mathrm{Sp}(X) \subset [0, \alpha[$, U is unitary and P is a projection. Consequently

$$\begin{aligned} \begin{pmatrix} f(\lambda x+(1-\lambda)y) & 0 \\ 0 & f(0) \end{pmatrix} &= f(PU^*XUP) \leqq Pf(U^*XU)P \\ &= PU^*f(X)UP = \begin{pmatrix} \lambda f(x)+(1-\lambda)f(y) & 0 \\ 0 & 0 \end{pmatrix}. \end{aligned}$$

Thus $f(\lambda x+(1-\lambda)y) \leqq \lambda f(x)+(1-\lambda)f(y)$ and $f(0) \leqq 0$.

2.2. Corollary. *Let $\pi: A \to B(H)$ be a positive, linear contraction on a C^*-algebra A. For every continuous operator convex function f defined on $[0, \alpha[$ with $f(0) \leqq 0$, and each self-adjoint x in A with spectrum in $[0, \alpha[$, we have*

$$f(\pi(x)) \leqq \pi(f(x)).$$

Proof. Restricting π to the commutative C^*-algebra generated by x, we may assume that π is completely positive. Using Stinespring's decomposition of π the result follows from Theorem 2.1 exactly as in the proof of [7, Corollary].

2.3. Remark. The result above is proved in [4, 3] under the additional hypothesis that $\pi(1)=1$. Note that with $f(t)=t^2$ we obtain Kadison's generalization of the Cauchy-Schwarz inequality: $\pi(x)^2 \leqq \pi(x^2)$. This is fitting, since Jensen himself first used his inequality to prove Cauchy's inequality, cf. [8, 9].

2.4. Theorem. *A continuous real function f on the interval $[0, \alpha[$ is operator convex with $f(0) \leqq 0$, if and only if*

(v) $\qquad t \to t^{-1}f(t)$ *is operator monotone on* $]0, \alpha[$.

Proof (ii)⇒(v): Suppose that x, y are operators with spectra in $]0, \alpha[$ and $x \leqq y$, and put $a=y^{-1/2}x^{1/2}$. Then $\|a\| \leqq 1$ so that by condition (ii) in Theorem 2.1

$$f(x)=f(a^*ya) \leqq a^*f(y)a = x^{1/2}y^{-1}f(y)x^{1/2}.$$

Multiplying left and right with $x^{-1/2}$ gives

$$x^{-1}f(x) \leqq y^{-1}f(y)$$

as desired.

(v)⇒(iv): Since f is continuous it suffices to prove 2.1 (iv) when x is invertible (replacing x with $x+\delta$ if necessary). Moreover, since $\|x\|<\alpha$ there is an $\varepsilon>0$ such that $\|(1+\varepsilon)x\|<\alpha$. Now $x^{1/2}(\varepsilon+p)x^{1/2} \leqq (1+\varepsilon)x$, whence by assumption

$$x^{-1/2}(\varepsilon+p)^{-1}x^{-1/2}f(x^{1/2}(\varepsilon+p)x^{1/2}) \leqq (1+\varepsilon)^{-1}x^{-1}f((1+\varepsilon)x).$$

Multiplication left and right with $px(\varepsilon+p)x^{1/2}$ and $x^{1/2}(\varepsilon+p)xp$ preserves the order and yields

$$px^{1/2}f(x^{1/2}(\varepsilon+p)x^{1/2})x^{1/2}(\varepsilon+p)xp \leqq (1+\varepsilon)^{-1}px(\varepsilon+p)f((1+\varepsilon)x)(\varepsilon+p)xp.$$

Since f is continuous we may let $\varepsilon\to 0$ to obtain

$$px^{1/2}f(x^{1/2}px^{1/2})x^{1/2}pxp \leqq pxp\,(pf(x)p)pxp\,.$$

Now $f(x^{1/2}px^{1/2})x^{1/2}p = x^{1/2}pf(pxp)$, because this is true for any polynomial f and therefore, by Weierstrass' approximation theorem, for any continuous function f. Consequently

$$pxp\;f(pxp)\,pxp \leqq pxp\,(pf(x)p)pxp\,,$$

from which we conclude that $pf(pxp)p \leqq pf(x)p$.

The function $t\to t^{-1}f(t)$ being increasing on $]0,\alpha[$ implies that $f(0)\leqq 0$. Thus with $g(t)=f(t)-f(0)$,

$$\begin{aligned} f(pxp) &= g(pxp)+f(0) \leqq g(pxp)+f(0)p \\ &= p(g(pxp)+f(0))p = pf(pxp)p\,. \end{aligned}$$

This in conjunction with the previous inequality gives the required result.

2.5. Theorem. *If $f\leqq 0$ is a continuous real function on the half-line $[0,\infty[$, the five conditions in Theorems 2.1 and 2.4 are again equivalent to the condition*

(vi) $\qquad -f$ *is operator monotone.*

Proof. (vi)⇒(ii): In essence, this is [7, Theorem]. But since the argument in [7] tacitly assumes that $f(0)=0$ we sketch the proof:

With notations as in the implication (i)⇒(ii) in Theorem 2.1 one shows [7] that for any $\varepsilon>0$ there is a sufficiently large λ such that

$$U^*XU = \begin{pmatrix} a^*xa & a^*xb \\ bxa & bxb \end{pmatrix} \leqq \begin{pmatrix} a^*xa+\varepsilon & 0 \\ 0 & \lambda \end{pmatrix} = Y.$$

Consequently,

$$\begin{aligned} \begin{pmatrix} a^*f(x)a & a^*f(x)b \\ bf(x)a & bf(x)b \end{pmatrix} &= U^*\begin{pmatrix} f(x) & 0 \\ 0 & 0 \end{pmatrix}U \geqq U^*\begin{pmatrix} f(x) & 0 \\ 0 & f(0) \end{pmatrix}U \\ &= U^*f(X)U = f(U^*XU) \geqq f(Y). \end{aligned}$$

As $\varepsilon\to 0$ this implies that $a^*f(x)a \geqq f(a^*xa)$.

(i)⇒(vi): If $0 \leqq x \leqq y$, take λ in $]0,1[$ and write

$$\lambda y = \lambda x + (1-\lambda)(\lambda(1-\lambda)^{-1}(y-x)).$$

This gives

$$f(\lambda y) \leqq \lambda f(x) + (1-\lambda) f(\lambda(1-\lambda)^{-1}(y-x)) \leqq \lambda f(x),$$

since $f \leqq 0$. Letting $\lambda \to 1$ we obtain $f(y) \leqq f(x)$.

2.6. Corollary. *A strictly positive, continuous function f on $]0,\infty[$ is operator monotone if and only if the function $t \to tf(t)^{-1}$ is operator monotone.*

Proof. The involution $t \to -t^{-1}$ is operator monotone on $]0,\infty[$ by Theorem 2.4. Considering the function $f_\varepsilon(t) = f(\varepsilon + t)$ on $[0,\infty[$, we see from the biimplication (vi)⇔(v) in Theorems 2.4 and 2.5 that $t \to -t^{-1} f_\varepsilon(t)$ is operator monotone. Applying the involution and letting $\varepsilon \to 0$ it follows that $t \to tf(t)^{-1}$ is operator monotone on $]0,\infty[$. For the converse, we know that $t \to (t+\varepsilon) f_\varepsilon(t)^{-1}$ is operator monotone on $[0,\infty[$. Reading the biimplications backwards it follows that $t \to t(t+\varepsilon)^{-1} f_\varepsilon(t)$ is operator monotone on $]0,\infty[$; whence f is operator monotone on $]0,\infty[$.

2.7. Remark. Some of the implications linking the six conditions in Theorems 2.1, 2.4, and 2.5 are known. Thus the biimplication (i)⇔(iv) is due to Davis [4], and the biimplication (i)⇔(v) to Bendat and Sherman [2]. Especially in the last case our proof, passing via the Jensen inequality, is simpler.

3. Differential Characterizations

Consider an open interval I in $\mathbb{R}$ and a continuously differentiable function f on I. Fix n and take $x = x^*$ in $\mathbb{M}_n$ with $\mathrm{Sp}(x) \subset I$. If $\{e_{ij} | 1 \leqq i, j \leqq n\}$ is a system of matrix units for $\mathbb{M}_n$ such that $x = \sum \lambda_i e_{ii}$, we shall denote by $f^{[1]}(x)$ the element in $\mathbb{M}_n$ with

$$f^{[1]}(x)_{ij} = (\lambda_i - \lambda_j)^{-1}(f(\lambda_i) - f(\lambda_j)) \quad \text{if} \quad \lambda_i \neq \lambda_j,$$
$$f^{[1]}(x)_{ij} = f'(\lambda_i) \quad \text{if} \quad \lambda_i = \lambda_j.$$

3.1. Lemma (cf. [5, III]). *With f and x as above we have*

$$\mathrm{Lim}\, \varepsilon^{-1}(f(x+\varepsilon h) - f(x)) = f^{[1]}(x) \underset{x}{\cdot} h$$

for every self-adjoint h in $\mathbb{M}_n$, where $\underset{x}{\cdot}$ denotes the Hadamard product of matrices in a basis that diagonalizes x.

Proof. It suffices to consider functions of the form $f(t) = t^p$, $p \geqq 0$, since their linear span is dense in $C^1(I)$. But

$$\varepsilon^{-1}((x+\varepsilon h)^p - x^p) = \sum_0^{p-1} x^k h x^{p-1-k} + \mathcal{O}(\varepsilon),$$

where $\mathcal{O}(\varepsilon) \to 0$ as $\varepsilon \to 0$. Consequently,

$$e_{ii}\, \mathrm{Lim}\, \varepsilon^{-1}((x+\varepsilon h)^p - x^p) e_{jj} = \sum_0^{p-1} \lambda_i^k \lambda_j^{p-1-k} e_{ii} h e_{jj}$$
$$= (\lambda_i - \lambda_j)^{-1}(\lambda_i^p - \lambda_j^p) h_{ij} e_{ij} = f^{[1]}(x)_{ij} h_{ij} e_{ij},$$

as desired.

3.2. Remark. The element $f^{[1]}(x)$ in many ways resembles the ordinary differential. The weakness of it is, of course, that the Hadamard product depends on the basis, so that neither the formulation nor the proof of, say, the mean value theorem is obvious. For our purposes the next result replaces the mean value theorem.

3.3. Lemma. *If $t \to x(t)$ is a C^1-function from $[0,1]$ to the space of self-adjoint $n \times n$-matrices with spectra in the open interval I, and if $f \in C^1(I)$, then*

$$f(x(1)) - f(x(0)) = \int_0^1 f^{[1]}(x(t))_{\dot{x}(t)}\, x'(t)\, dt.$$

Proof. The argument in the proof of Lemma 3.1 shows that

$$f(x+h) - f(x) = f^{[1]}(x)_{\dot{x}}\, h + \|h\|\, \mathcal{O}(h),$$

where $\mathcal{O}(h) \to 0$ as $h \to 0$. To every pair s, t in $[0,1]$ we can therefore find a matrix $\mathcal{O}(s,t)$ such that

$$f(x(s)) - f(x(t)) = f^{[1]}(x(t))_{\dot{x}(t)}\,(x(s) - x(t)) + \|x(s) - x(t)\|\, \mathcal{O}(s,t);$$

and for a given $\varepsilon > 0$ we can find n in $\mathbb{N}$ such that $\|\mathcal{O}(s,t)\| < \varepsilon$ whenever $|s-t| \leqq n^{-1}$. With $x_k = x(kn^{-1})$, $0 \leqq k \leqq n$, we therefore have

$$\begin{aligned} f(x(1)) - f(x(0)) &= \sum_{k=0}^{n-1} f(x_{k+1}) - f(x_k) \\ &= \sum_{k=0}^{n-1} f^{[1]}(x_k)_{\dot{x}_k}\,(x_{k+1} - x_k) \\ &\quad + \|x_{k+1} - x_k\|\, \mathcal{O}((k+1)n^{-1}, kn^{-1}). \end{aligned}$$

The first part of the sum converges to the Riemann integral

$$\int_0^1 f^{[1]}(x(t))_{\dot{x}(t)}\, x'(t)\, dt$$

as $n \to \infty$. The second is bounded by

$$\sum_{k=0}^{n-1} n^{-1}\, \|x'(\theta_k)\|\, \varepsilon \leqq \varepsilon \operatorname{Sup} \|x'(t)\|.$$

The desired formula follows.

3.4. Proposition. *A C^1-function f on an open interval I is operator monotone if and only if $f^{[1]}(x) \geqq 0$, for every self-adjoint x in $\mathbb{M}_n$ with $\operatorname{Sp}(x) \subset I$, and every n in $\mathbb{N}$.*

Proof. If f is operator monotone and $x = \sum \lambda_i e_{ii}$, take $h = \sum e_{ij}$. Then $n^{-1}h$ is a (one-dimensional) projection; in particular $h \geqq 0$. Consequently

$$0 \leqq \varepsilon^{-1}(f(x + \varepsilon h) - f(x)),$$

whence $f^{[1]}(x) \geqq 0$ by Lemma 3.1, since h is the unit for the Hadamard product.

Conversely, if x and y have spectra in I and $x \leqq y$, set $x(t)=(1-t)x+ty$. Then $x'(t)=y-x \geqq 0$, and thus by Lemma 3.3

$$f(y)-f(x)=\int_0^1 f^{[1]}(x(t))_{\dot{x}(t)}(y-x)dt \geqq 0,$$

because the Hadamard product of positive matrices is again positive.

3.5. Lemma. *If a continuous function f is operator monotone on $]-1,1[$, then $t \to (t+\lambda)f(t)$ is operator convex for every λ, $|\lambda| \leqq 1$.*

Proof. Assume first that f is operator monotone on $[-1,1]$. Then the function $g(t)=tf(t-1)$ is operator convex on $[0,2[$ by Theorem 2.4, since $t \to t^{-1}g(t)$ is operator monotone and $g(0)=0$. Thus the function $t \to (t+1)f(t)$ $(=g(t+1))$ is operator convex on $[-1,1[$. Applying this result to the operator monotone function $t \to -f(-t)$ we find that $t \to -(t+1)f(-t)$ is operator convex. But the transformation $t \to -t$ preserves convexity, and thus $t \to (t-1)f(t)$ is operator convex. Consequently the function

$$t \to (t+\lambda)f(t)=\tfrac{1}{2}(1+\lambda)(t+1)f(t)+(1-\tfrac{1}{2}(1+\lambda))(t-1)f(t)$$

is operator convex.

In the general case use the result above to conclude that the function $t \to (t+\lambda)f((1-\varepsilon)t)$ is operator convex on $]-1,1[$ for every $\varepsilon>0$, hence also for $\varepsilon=0$.

3.6. Theorem. *Every operator monotone function f on $]-1,1[$ is continuously differentiable.*

Proof. Choose a symmetric C^∞-function $\varphi \geqq 0$ with support in $[-1,1]$ and integral 1. For $0<\varepsilon<1$ and $|t|<1-\varepsilon$ put

$$f_\varepsilon(t)=\varepsilon^{-1}\int_{-\varepsilon}^{\varepsilon} f(t-s)\varphi(\varepsilon^{-1}s)ds.$$

Note that f_ε is operator monotone and belongs to $C^\infty(]\varepsilon-1,\ 1-\varepsilon[)$. Moreover, since f has limits from left and from right at every point t in $]-1,1[$, we have

$$\operatorname*{Lim}_{\varepsilon} f_\varepsilon(t)=\tilde{f}(t)=\tfrac{1}{2}(f(t+)+f(t-)).$$

Since f_ε is continuous, the function $g_\varepsilon(t)=(t+1)f_\varepsilon(t)$ is (operator) convex by Lemma 3.5. Consequently the limit function $\tilde{g}(t)=(t+1)\tilde{f}(t)$ is (operator) convex and therefore continuous on $]-1,1[$. Since $t+1 \quad 0$ on $]-1,1[$ we conclude that $\tilde{f}$ is continuous, which implies that $\tilde{f}=f$.

Every convex function g on $]-1,1[$ has left and right derivatives $g'(t-)$ and $g'(t+)$ at every point t and these satisfy the relations

$$(*) \qquad g'(t-) \leqq g'(t+), \qquad g'(t+\delta)\pm) \to g'(t+), \qquad g'((t-\delta)\pm) \to g'(t-),$$

as $\delta \searrow 0$. With $g(t)=(t+1)f(t)$ we find that $g'(t\pm)=f(t)+(t+1)f'(t\pm)$, and since $t+1>0$ for all t in $]-1,1[$, the derivatives of f satisfy the exact same relations $(*)$ as those of g. For s, t in $]-1,1[$ consider the 2×2-matrix $x=se_{11}+te_{22}$.

Eventually s, t belong to $]\varepsilon-1, 1-\varepsilon[$ and since $f_\varepsilon \in C^1(]\varepsilon-1, 1-\varepsilon[)$ it follows from Proposition 3.4 that $f_\varepsilon^{[1]}(x) \geqq 0$; in particular the determinant is positive, i.e.

$$(s-t)^{-2}(f_\varepsilon(s)-f_\varepsilon(t))^2 \leqq f_\varepsilon'(s) f_\varepsilon'(t).$$

From the first part of the proof we know that f is continuous, so $f_\varepsilon \to f$ uniformly on compact subsets of $]-1, 1[$. Moreover, $f_\varepsilon'(t) \to \frac{1}{2}(f'(t+)+f'(t-))$. It follows that

$$(s-t)^{-2}(f(s)-f(t))^2 \leqq \tfrac{1}{4}(f'(s+)+f'(s-))(f'(t+)+f'(t-)).$$

with $s \searrow t$ this implies by $(*)$ that

$$f'(t+)^2 \leqq \tfrac{1}{4}(f'(t+)+f'(t+))(f'(t+)+f'(t-)),$$

and consequently $f'(t+)=f'(t-)$. Thus f is differentiable, and $(*)$ shows that f' is continuous.

3.7. Lemma (cf. [5, III]). *If $f \in C^2(]-1, 1[)$ with $f(0)=0$, and $\{e_{ij} | 0 \leqq i, j \leqq n\}$ is a system of matrix units for $\mathbb{M}_{n+1}$, then with*

$$h=\sum_{i=1}^{n} e_{i0}+e_{0i}, \qquad q=1-e_{00}, \qquad x=\sum_{i=1}^{n} \lambda_i e_{ii},$$

where $|\lambda_i|<1$, we have

$$\operatorname*{Lim}_{\varepsilon} q\varepsilon^{-2}(f(x+\varepsilon h)-f(x)-\varepsilon f^{[1]}(x) \underset{x}{\cdot} h)q = g^{[1]}(x),$$

where $g(t)=t^{-1}f(t)$ (and $g(0)=f'(0)$).

Proof. If $f(t)=t^p$, then the second order term in ε of $f(x+\varepsilon h)$ is $b=\sum x^m h x^r h x^s$, the summation being over all integers $m, r, s \geqq 0$ such that $m+r+s=p-2$. However, $xhx=0$ by our choice of h, whence

$$\begin{aligned} qbq &= q(hx^{p-2}h)q+q(\textstyle\sum x^m h^2 x^{p-2-m})q \\ &= q(\textstyle\sum x^m(\sum e_{ij}+e_{0i}e_{j0})x^{p-2-m})q \\ &= \textstyle\sum x^m e_{ij} x^{p-2-m} = \sum \lambda_i^m \lambda_j^{p-2-m} e_{ij} \\ &= \textstyle\sum (\lambda_i-\lambda_j)^{-1}(\lambda_i^{p-1}-\lambda_j^{p-1}) e_{ij} = g^{[1]}(x). \end{aligned}$$

The general case follows by approximation exactly as in Lemma 3.1.

3.8. Lemma. *If $f \in C^2(]-1, 1[)$ with $f(0)=0$ and f is operator convex, then the function $g: t \to t^{-1}f(t)$ is operator monotone.*

Proof. If $x \in \mathbb{M}_n$ and $x=x^*$ with $\mathrm{Sp}(x) \subset]-1, 1[$, then

$$f(x+\delta\varepsilon h)=f((1-\delta)x+\delta(x+\varepsilon h)) \leqq (1-\delta)f(x)+\delta f(x+\varepsilon h).$$

Dividing by δ, and then letting $\delta \to 0$ gives

$$0 \leqq f(x+\varepsilon h)-f(x)-\varepsilon f^{[1]}(x) \underset{x}{\cdot} h.$$

With h as in Lemma 3.7 it follows from Lemma 3.7 that $g^{[1]}(x) \geqq 0$. This holds for all x as above, whence g is operator monotone by Proposition 3.4.

3.9. Theorem. *If f is operator monotone on $]-1,1[$ and $f(0)=0$, then the function $t\to(1+\lambda t^{-1})f(t)$ is operator monotone for $|\lambda|\leqq 1$.*

Proof. Assume first that $f\in C^2(]-1,1[)$. By Lemma 3.5 the function $t\to(t+\lambda)f(t)$ is operator convex and by Lemma 3.8 this implies that $t\to(1+\lambda t^{-1})f(t)$ is operator monotone.

In the general case define f_ε as in the proof of Theorem 3.6. Since $f\in C^1(]-1,1[)$ by Theorem 3.6, we know that $f_\varepsilon\to f$ and $f_\varepsilon'\to f'$ uniformly on compact subsets of $]-1,1[$. In particular

$$(1+\lambda t^{-1})(f_\varepsilon(t)-f_\varepsilon(0))\to(1+\lambda t^{-1})f(t)$$

for every t, which implies that the limit function is operator monotone.

3.10. Corollary. *If f is operator monotone on $]-1,1[$ and $f(0)=0$, then $f''(0)$ exists.*

Proof. The function $t\to(1+t^{-1})f(t)$ is continuously differentiable by Theorems 3.6 and 3.9. Its derivative at 0 is $f'(0)+\frac{1}{2}f''(0)$.

3.11. Remarks. Except for the order in which the results are established, little in this section is new. The characterization of operator monotonicity by positive (generalized) differentials is very close to Löwner's original characterization [11, p. 183]. The proof used here for the necessity of the condition goes back to Daleckij and Krein; see the discussion in [5, III]. We noticed that the sufficiency of the condition (i.e., $f^{[1]}\geqq 0$) can be obtained quite easily by integrating the differential. Unlike Löwner's original characterization the one we use (Proposition 3.4) requires the function to be differentiable; and thus it becomes necessary to establish separately that operator monotone functions are continuously differentiable (Theorem 3.6). Our proof of Theorem 3.6 uses the same smoothing technique as Nagy in [12], but is simplified by use of the Jensen operator inequality. Of course, the main point of this section is to establish Theorem 3.9 (due to Bendat and Sherman) as a prelude to Löwner's theorem, not a corollary.

4. Löwner's Theorem

Throughout this section we shall denote by K the set of operator monotone functions f on $]-1,1[$, such that $f(0)=0$ and $f'(0)=1$.

4.1. Lemma. *If $f\in K$ then*

$$f(t)\leqq t(1-t)^{-1} \quad \textit{for} \quad t\geqq 0;$$
$$f(t)\geqq t(1+t)^{-1} \quad \textit{for} \quad t\leqq 0;$$
$$|f''(0)|\leqq 2.$$

Proof. Consider the 2×2-matrix $x=te_{11}+0e_{22}$. By Proposition 3.4, $f^{[1]}(x)\geqq 0$, whence

$$(*) \qquad t^{-2}f(t)^2\leqq f'(t)f'(0)=f'(t).$$

By Lemma 3.5 the two functions $t \to (t \pm 1) f(t)$ are (operator) convex. Their derivatives (cf. Theorem 3.6) are therefore increasing. For $t > 0$ this implies that

$$(**) \qquad f(t) + (t-1) f'(t) \geqq -1;$$

whereas for $t < 0$ we may conclude that

$$(***) \qquad f(t) + (t+1) f'(t) \leqq 1.$$

Inserting $(*)$ in $(**)$ we obtain for all $t > 0$,

$$f(t) + 1 \geqq (1-t) t^{-2} f(t)^2.$$

If now $f(t) > t(1-t)^{-1}$ for some $t > 0$ then, multiplying with $f(t)$ and inserting the result in the inequality above, we get $f(t) + 1 > t^{-1} f(t)$; whence $f(t) < t(1-t)^{-1}$, a contradiction. Therefore $f(t) \leqq t(1-t)^{-1}$ for all $t \geqq 0$. The other inequality is obtained similarly by inserting $(*)$ in $(***)$.

Finally, as in the proof of Corollary 3.10,

$$\begin{aligned} f'(0) + \tfrac{1}{2} f''(0) &= \operatorname{Lim} t^{-1}((1+t^{-1}) f(t) - f'(0)) \\ &\leqq \operatorname{Lim} t^{-1}((1+t^{-1}) t(1-t)^{-1} - 1) = 2, \end{aligned}$$

as $t \searrow 0$; whereas for $t \nearrow 0$ we have

$$f'(0) + \tfrac{1}{2} f''(0) \geqq \operatorname{Lim} t^{-1}((1+t^{-1}) t(1+t)^{-1} - 1) = 0.$$

Combining the two estimates we have $|f''(0)| \leqq 2$.

4.2. Lemma. *The set K is convex and compact in the topology of pointwise convergence.*

Proof. Convexity is clear. Take now a universal net $\{f_i\}$ in K. It follows from Lemma 4.1 that each net $\{f_i(t)\}$ is bounded, so that $\{f_i\}$ is pointwise convergent to a function f. Therefore f is operator monotone on $]-1, 1[$ and $f(0) = 0$.

The functions $(1+t^{-1}) f_i(t)$ are (operator) monotone by Theorem 3.9. Consequently

$$\begin{aligned} (1-\varepsilon^{-1}) f(-\varepsilon) &= \operatorname{Lim}(1-\varepsilon^{-1}) f_i(-\varepsilon) \leqq \operatorname{Lim} f_i'(0) = 1; \\ (1+\varepsilon^{-1}) f(\varepsilon) &= \operatorname{Lim}(1+\varepsilon^{-1}) f_i(\varepsilon) \geqq \operatorname{Lim} f_i'(0) = 1. \end{aligned}$$

This holds for every $\varepsilon > 0$, and since $f \in C^1(]-1, 1[)$ we conclude that $f'(0) = 1$.

4.3. Proposition. *The extreme points in K have the form*

$$f(t) = t(1-\alpha t)^{-1}, \quad \textit{where} \quad \alpha = \tfrac{1}{2} f''(0).$$

Proof. For any f in K define

$$g_\lambda(t) = (1 + \tfrac{1}{2}\lambda f''(0))^{-1}((1+\lambda t^{-1}) f(t) - \lambda)$$

for $|\lambda| < 1$. Note that $f''(0)$ exists by Corollary 3.10 and that $|\frac{1}{2}\lambda f''(0)| < 1$ by Lemma 4.1. It follows from Theorem 3.9 that g_λ is operator monotone. In fact the constants have been chosen so that $g_\lambda \in K$. It is immediate that

$$f = \tfrac{1}{2}(1 + \tfrac{1}{2}\lambda f''(0)) g_\lambda + \tfrac{1}{2}(1 - \tfrac{1}{2}\lambda f''(0)) g_{-\lambda}.$$

If therefore f is extreme, then $f=g_\lambda$; or

$$(1+\tfrac{1}{2}\lambda f''(0))f(t)=(1+\lambda t^{-1})f(t)-\lambda,$$

from which we immediately see that $f(t)=t(1-\frac{1}{2}f''(0)t)^{-1}$.

4.4. Theorem. *For each non-constant operator monotone function f on $]-1,1[$ there is a unique probability measure μ on $[-1,1]$ such that*

$$f(t)=f(0)+f'(0)\int t(1-\alpha t)^{-1}\,d\mu(\alpha).$$

Proof. Without loss of generality we may assume that $f\in K$ [for if $f'(0)=0$, then f is constant by $(*)$ in the proof of 4.1]. Put $f_\alpha(t)=t(1-\alpha t)^{-1}$ for $-1\leqq\alpha\leqq 1$. Since K is convex and compact by Lemma 4.2 and $\partial K\subset\{f_\alpha\}$ by Proposition 4.3, there is by Krein-Milman's theorem a net $\{f_i\}$ in K converging to f, where $f_i=\int f_\alpha d\mu_i(\alpha)$ for each i, and μ_i is a probability measure on $[-1,1]$ with finite support. Since the space of probability measures is weak* compact the net $\{\mu_i\}$ has an accumulation point μ, and we see that

$$f(t)=\int f_\alpha(t)\,d\mu(\alpha)$$

for every t in $]-1,1[$.

From the expression above we see that f has an extension, necessarily unique, to a complex function

$$f(z)=\int z(1-\alpha z)^{-1}d\mu(\alpha)$$

which is holomorphic in $\mathbb{C}\backslash\mathbb{R}$. However, for every continuous function φ on $\mathbb{R}$ with compact support we have

$$\begin{aligned}&\operatorname*{Lim}_{\varepsilon}\pi^{-1}\int\varphi(s)\operatorname{Im}f((s-i\varepsilon)^{-1})ds\\&=\operatorname*{Lim}_{\varepsilon}\pi^{-1}\iint\varphi(s)((\alpha-s)^2+\varepsilon^2)^{-1}\varepsilon\,d\mu(\alpha)ds\\&=\operatorname*{Lim}_{\varepsilon}\int\varphi*h_\varepsilon(\alpha)d\mu(\alpha)=\int\varphi(\alpha)d\mu(\alpha),\end{aligned}$$

because $h_\varepsilon(s)=\pi^{-1}(s^2+\varepsilon^2)^{-1}\varepsilon$ is an approximate unit for $L^1(\mathbb{R})$. This proves that μ is unique. We learn thus, a posteriori, that K is a Bauer simplex with $\partial K=\{f_\alpha\}$.

4.5. Theorem. *For each non-linear operator convex function f on $]-1,1[$ there is a unique probability measure μ on $[-1,1]$ such that*

$$f(t)=f(0)+f'(0)t+\tfrac{1}{2}f''(0)\int t^2(1-\alpha t)^{-1}d\mu(\alpha).$$

Proof. By Theorem 2.4 the function

$$t\to(t+1)^{-1}(f((1-\varepsilon)t)-f(-1+\varepsilon))$$

is operator monotone on $]-1,1[$ for every $\varepsilon>0$. Thus $f\in C^2(]-1,1[)$ by Theorem 4.4. Assuming, as we may, that $f(0)=0$ and $f'(0)=0$ it follows from Lemma 3.8 in conjunction with Theorem 4.4 that

$$t^{-1}f(t)=s\int f_\alpha(t)\,d\mu(\alpha)$$

for some unique probability measure μ on $[-1,1]$, and $s=\frac{1}{2}f''(0)$.

4.6. Remark. An extremely elegant proof of Löwner's theorem was given by Korányi in [10]. It does, however, involve some delicate operator extension arguments. Having Theorem 3.9 at our disposal, these can be elimated as the following sketch of Korányi's proof will show:

Given an operator monotone function f in $C^1(]-1,1[)$ with $f(0)=0$, consider the subspace H_0 of $l^2(]-1,1[)$ consisting of functions of finite support. Let x denote the diagonal operator with respect to the natural basis, for which $x\delta_t=t\delta_t$. Now define a new inner product on H_0 by

$$\langle\xi|\eta\rangle=(f^{[1]}(x)\xi|\eta),$$

using Proposition 3.4 for the positivity of $\langle\cdot|\cdot\rangle$; and let H denote the Hilbert space completion of $H_0-\mathrm{Ker}\langle\cdot|\cdot\rangle$. Put $g(t)=t^{-1}f(t)$ [and $g(0)=f'(0)$] and define a symmetric bilinear form on H_0 by

$$\langle T_0\xi|\eta\rangle=(g^{[1]}(x)\xi|\eta).$$

The operator monotonicity of $f\pm g$ obtained in Theorem 3.9 shows that

$$\langle(1\pm T_0)\xi|\xi\rangle=((f\pm g)^{[1]}(x)\xi|\xi)\geqq 0,$$

so that T_0 extends by continuity to a self-adjoint operator T on H with $\|T\|\leqq 1$. For t in $]-1,1[$ let $\xi_0=(1-tT)\delta_t$ and compute that

$$\langle\xi_0|\xi\rangle=\langle(1-tT)\delta_t|\xi\rangle=\sum g(s)\bar{\xi}(s)$$

for every ξ in H_0. In particular, ξ_0 does not depend on t. Now let μ be the measure on $\mathrm{Sp}(T)$ ($\subset[-1,1]$) such that

$$(h(T)\xi_0|\xi_0)=\int h(\alpha)d\mu(\alpha)$$

for every continuous function h on $\mathrm{Sp}(T)$. Then

$$\begin{aligned}\int(1-\alpha t)^{-1}d\mu(\alpha)&=((1-tT)^{-1}\xi_0|\xi_0)\\&=(\delta_t|\xi_0)=g(t)=t^{-1}f(t).\end{aligned}$$

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Note added in proof. The implication (i)⇒(ii) in Theorem 2.1 is proved by D. Kainuma and M. Nakamura: Around Jensen's inequality. Math. Japonica **25**, 585–588 (1980). Their proof is based on the Kubo-Ando theory for means and requires Löwner's theorem. Along the same lines an alternate proof of the implication (vi)⇒(ii) in Theorem 2.5 has been given by J. I. Fujii.